

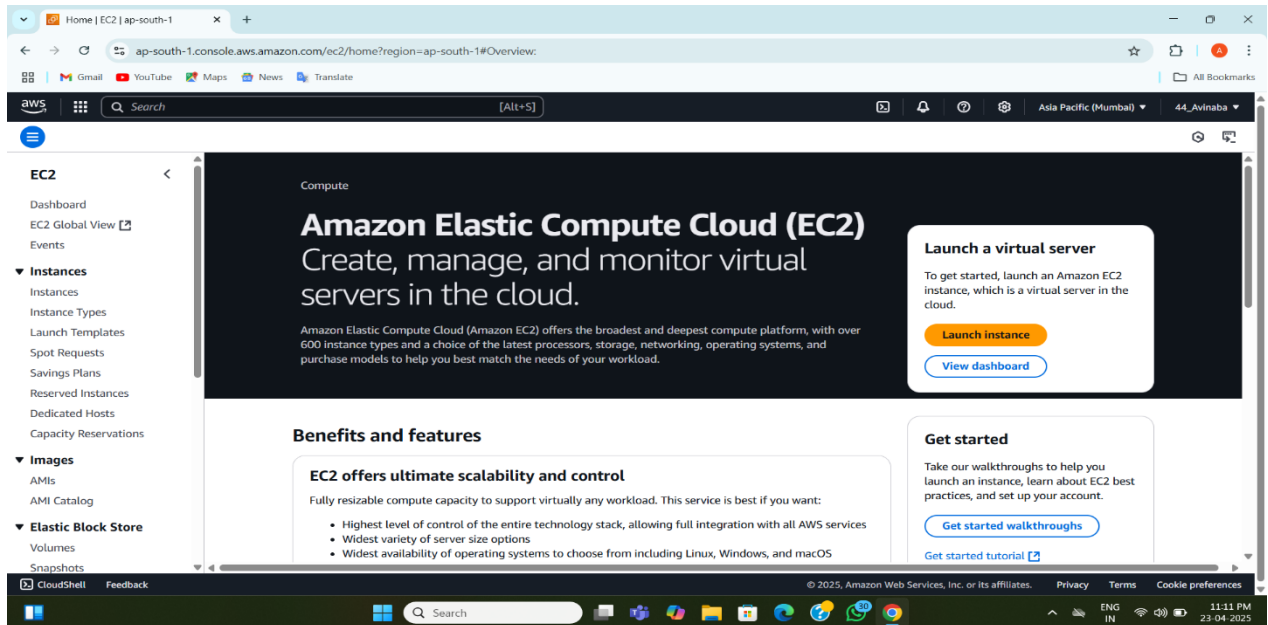
Assignment-9

PROBLEM STATEMENT :

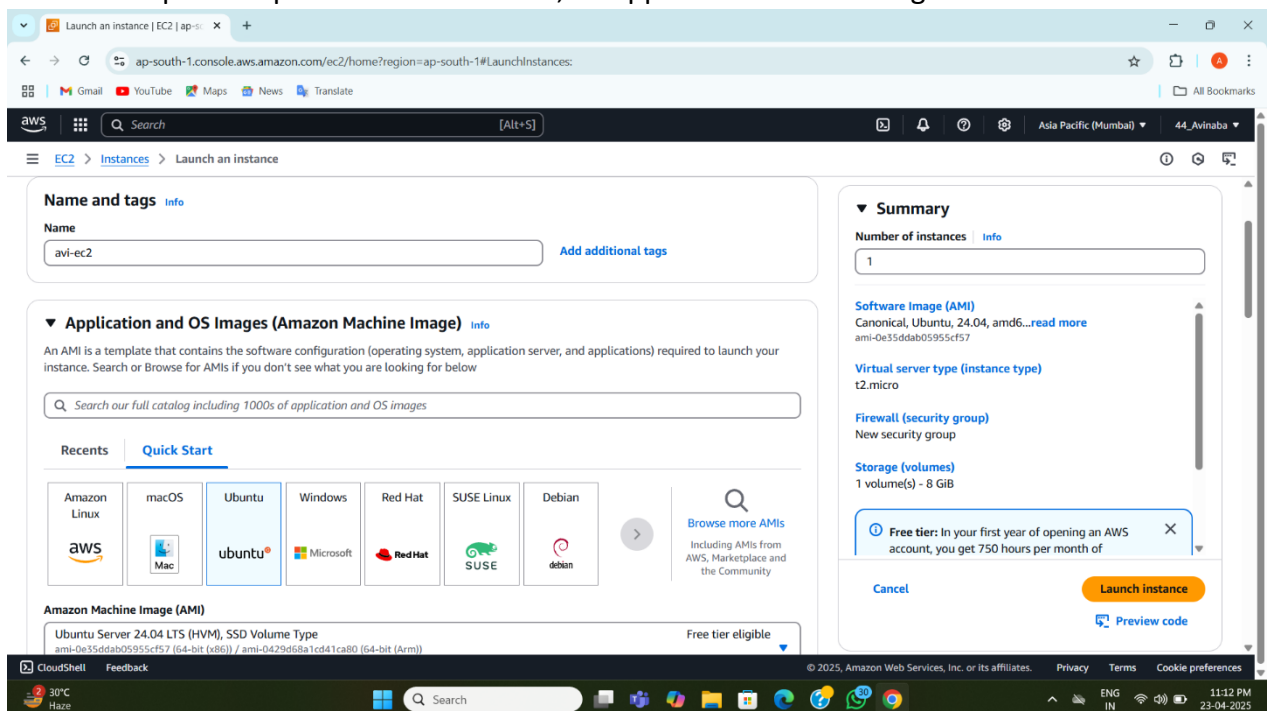
Deploy a project from GitHub to EC2.

Steps to deploy project :

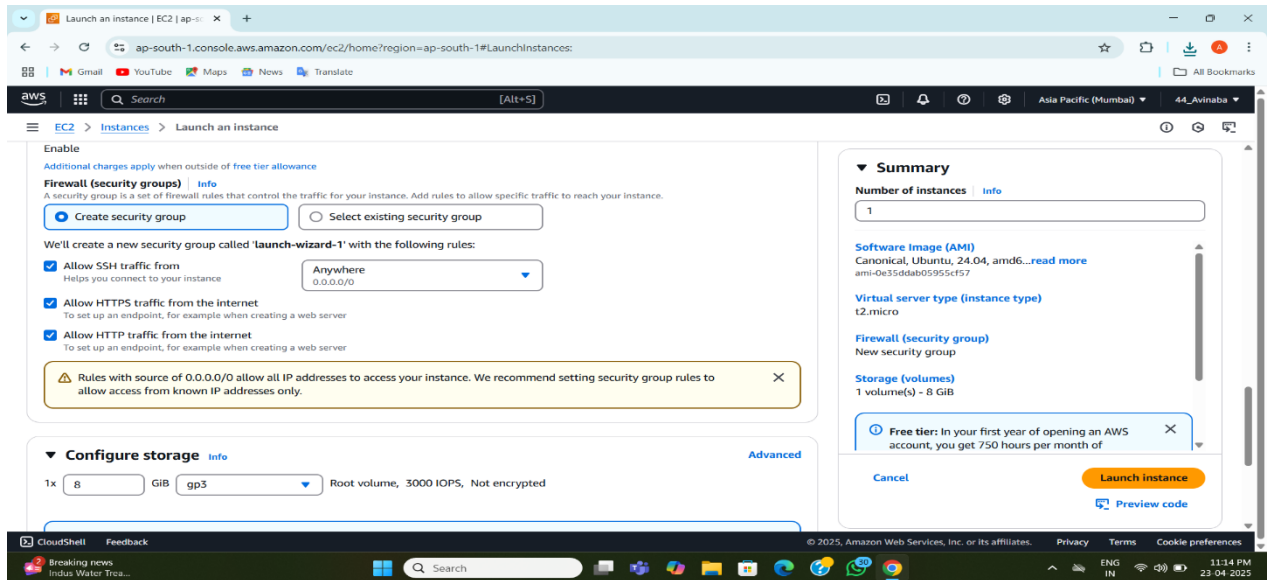
1. Sign up for an AWS account, search for 'EC2', select it and click on launch instance.



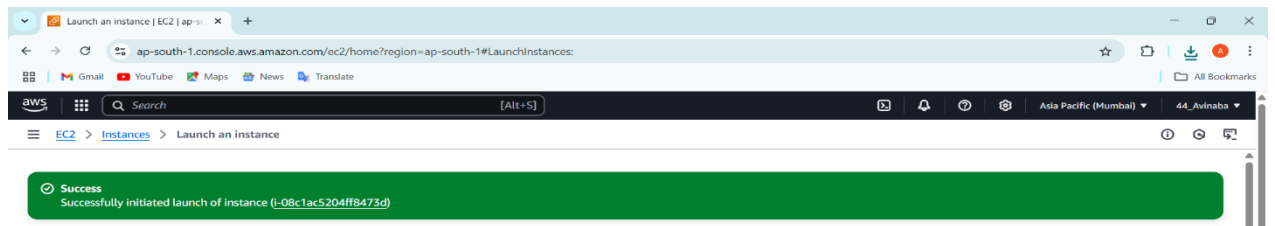
2. Fill up the required details->'Name', in 'Application and OS Images' select 'ubuntu'.



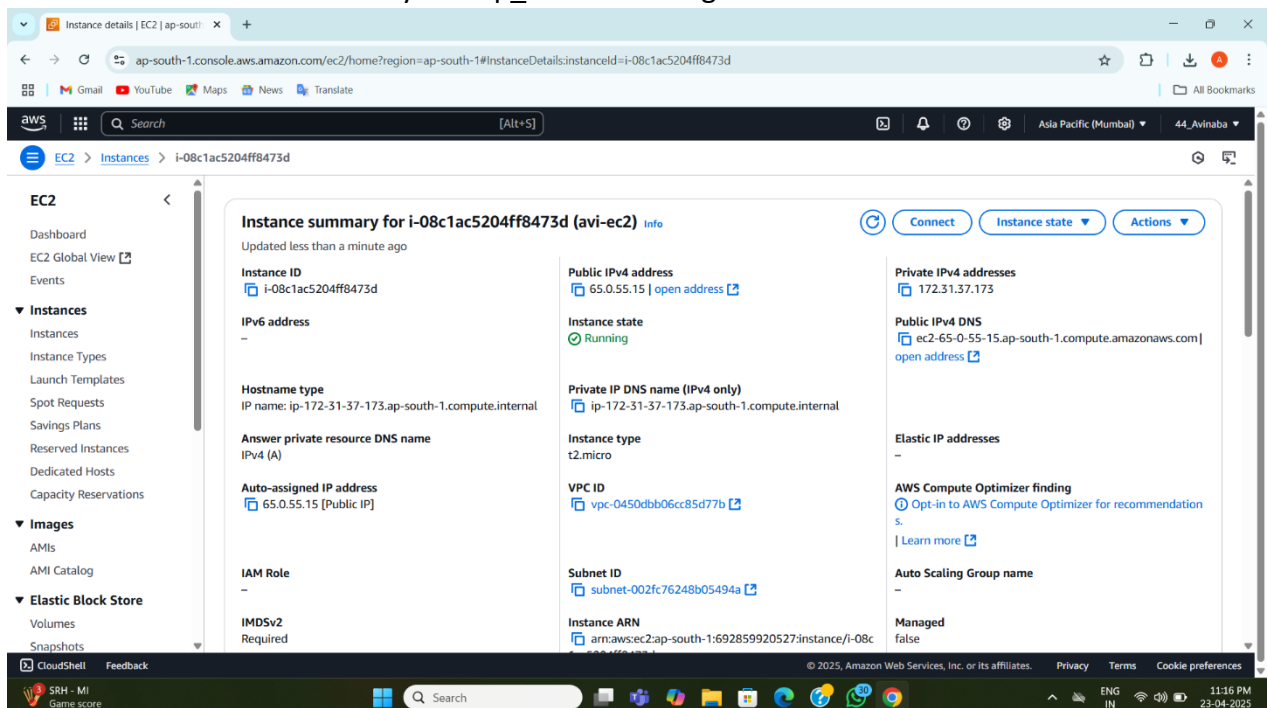
Under 'Network settings', tick off all the three checkboxes then click on 'Launch instance'.



3. "Dymanicp_EC2" instance is successfully created and then click on 'Instances'.



4. Now check whether "Dymanicp_EC2" is running or not then click on "Instance ID".



5. Then go to “Security” and click on link under “Security groups” to get the edit option.

The screenshot shows the AWS Management Console for an EC2 instance. The left sidebar contains navigation links for EC2, Images, and Elastic Block Store. The main content area is divided into tabs: Details, Status and alarms, Monitoring, Security, Networking, Storage, and Tags. The Security tab is active, displaying the following information:

- Security details:**
 - IAM Role:** -
 - Owner ID:** 692859920527
 - Launch time:** Wed Apr 23 2025 23:15:03 GMT+0530 (India Standard Time)
 - Security groups:** sg-071f74d1254e43d45 (launch-wizard-1)
- Inbound rules:** A table with 6 columns: Name, Security group rule ID, Port range, Protocol, Source, and Security group. It lists three rules:

Name	Security group rule ID	Port range	Protocol	Source	Security group
-	sgr-06382e7a0c521ec10	443	TCP	0.0.0.0/0	launch-wizard-
-	sgr-00018b8f3c0fca445	22	TCP	0.0.0.0/0	launch-wizard-
-	sgr-0c423e7051717ce12	80	TCP	0.0.0.0/0	launch-wizard-

6. Then in “Inbound rules”, click on “Edit inbound rules”.

This screenshot is identical to the previous one, showing the AWS Management Console for an EC2 instance with the Security tab active. It displays the same security details and inbound rules table.

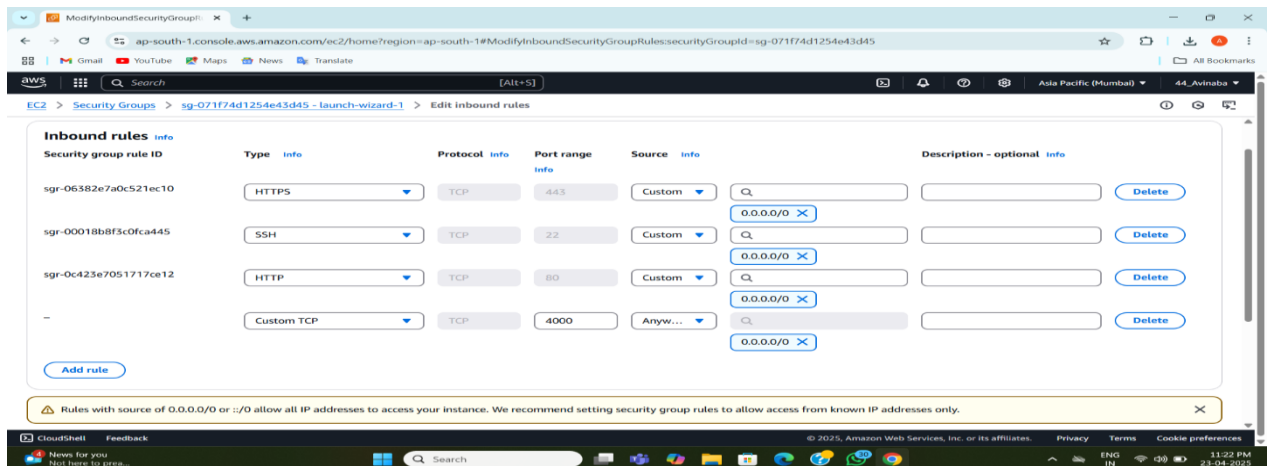
7. Click on “Add rule”.

The screenshot shows the “Edit inbound rules” page in the AWS Management Console. The breadcrumb trail indicates the path: EC2 > Security Groups > sg-071f74d1254e43d45 - launch-wizard-1 > Edit inbound rules. The page title is “Edit inbound rules” with an “Info” link. Below the title, it states: “Inbound rules control the incoming traffic that's allowed to reach the instance.”

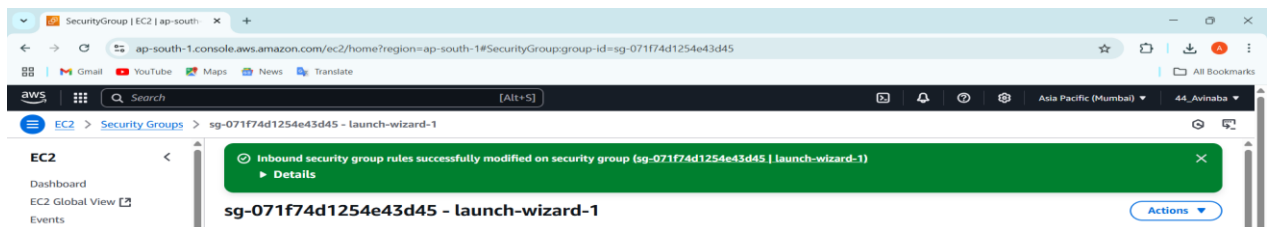
The main content area displays a table of inbound rules with columns: Security group rule ID, Type, Protocol, Port range, Source, and Description - optional. Each rule has a “Delete” button. At the bottom left, there is an “Add rule” button.

Security group rule ID	Type	Protocol	Port range	Source	Description - optional	Action
sgr-06382e7a0c521ec10	HTTPS	TCP	443	Custom	0.0.0.0/0	Delete
sgr-00018b8f3c0fca445	SSH	TCP	22	Custom	0.0.0.0/0	Delete
sgr-0c423e7051717ce12	HTTP	TCP	80	Custom	0.0.0.0/0	Delete
-	Custom TCP	TCP	0	Custom	0.0.0.0/0	Delete

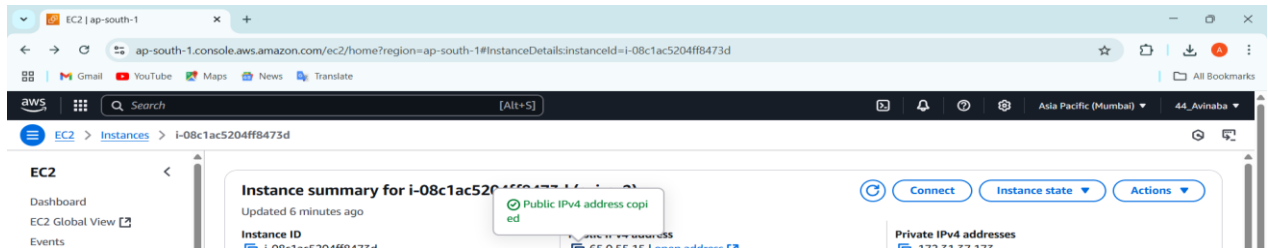
8. Set the “Port range” to 4000 , in “Source” set “0.0.0.0/0” & click on “Save rules”.



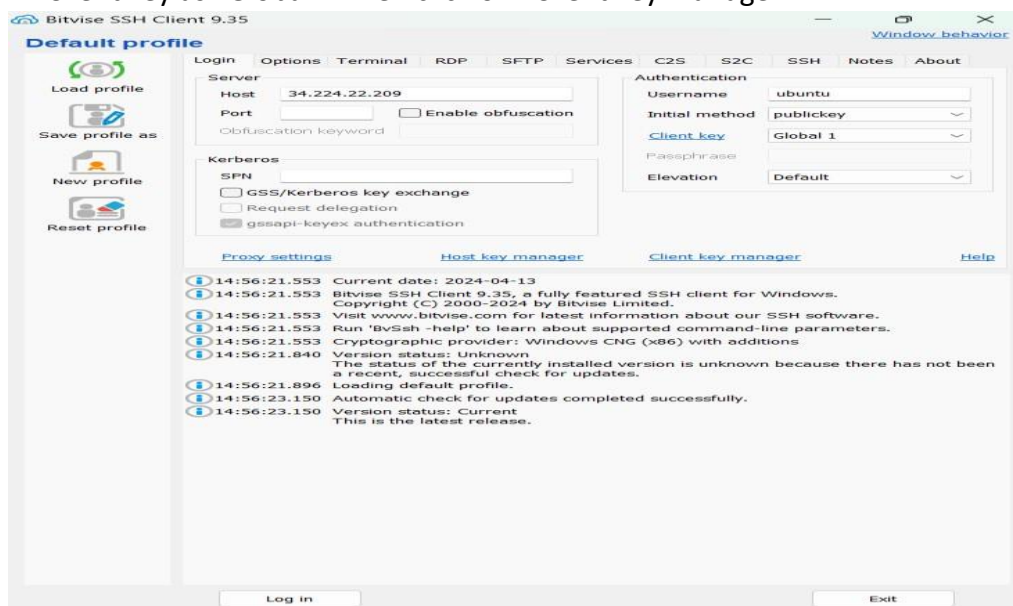
9. Rule has been successfully saved.



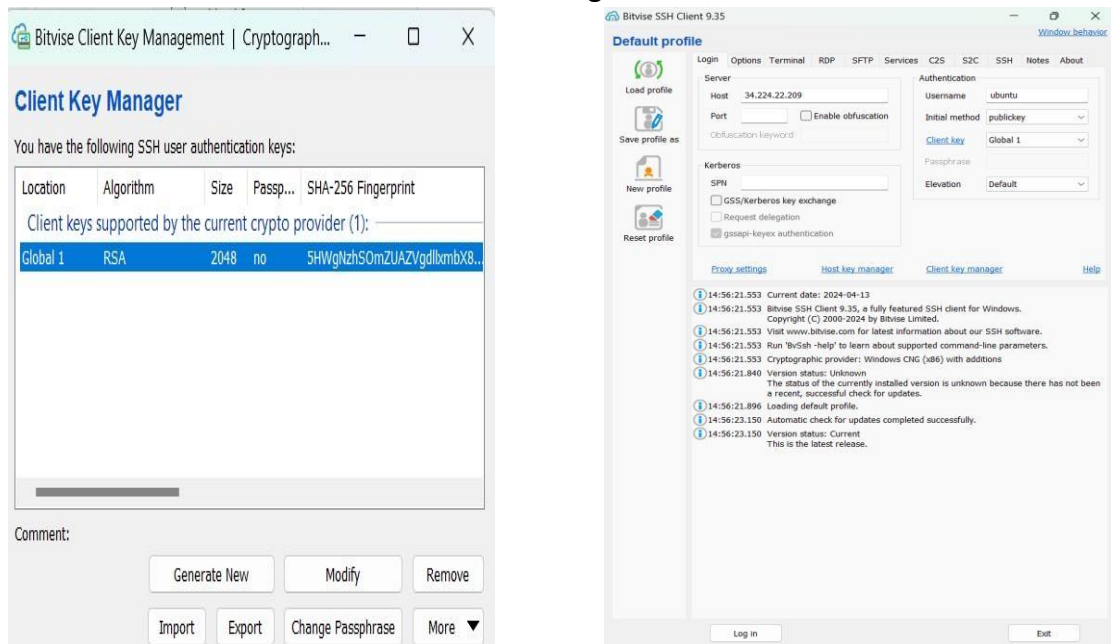
10. Go back to the Instance, copy the “Public IPv4 address”.



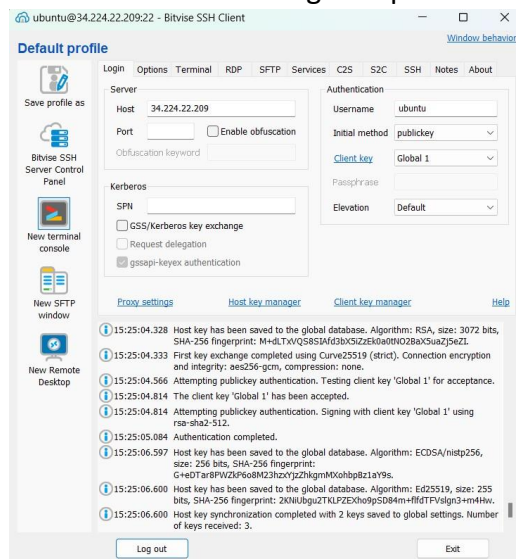
11. In “Bitwise SSH Client”, paste the “Public IPv4 address” in “Host” and under “Authentication tab” give the username as Ubuntu, Initial method as “Public key”, Client key as “Global1”.Then click on “Client key manager”.



12. Remove any previously selected key if any, then click on “Import” & select the key with which instance was created. Then click on Log in.



13. After successful “Log in” open a “New Terminal Console”.



14. In the console, type the following commands in sequential order.

```
ubuntu@34.224.22.209:22 - Bitvise xterm - ubuntu@ip-172-31-27-221: ~  
ubuntu@ip-172-31-27-221:~$ pwd  
/home/ubuntu  
ubuntu@ip-172-31-27-221:~$ sudo apt-get update  
Fetching 30.7 MB in 6s (5382 kB/s)  
Reading package lists... Done  
ubuntu@ip-172-31-27-221:~$ sudo apt-get upgrade  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
ubuntu@ip-172-31-27-221:~$ sudo apt-get install nginx
```

```

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-27-221:~$ nginx -v
nginx version: nginx/1.18.0 (Ubuntu)
ubuntu@ip-172-31-27-221:~$ curl -SL https://deb.nodesource.com/setup_16.x | sudo -E bash
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 19.6 kB  100 19.6 kB    0s 0.00s  0s 24.5 kB/s
Reading package lists... Done
2024-04-13 10:28:31 - Repository configured successfully. To install Node.js, run: apt-get install nodejs -y
ubuntu@ip-172-31-27-221:~$ sudo apt install nodejs

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-27-221:~$ nodejs -v
v16.20.2
ubuntu@ip-172-31-27-221:~$ git clone https://github.com/Rupsa1037/Rup_AWS1.git
Cloning into 'Rup_AWS1'...
remote: Enumerating objects: 7, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (7/7), done.
remote: Total 7 (delta 0), reused 7 (delta 0), pack-reused 0
Receiving objects: 100% (7/7), 48.21 KiB | 8.03 MiB/s, done.
ubuntu@ip-172-31-27-221:~$ ls
Rup_AWS1
ubuntu@ip-172-31-27-221:~$ cd Rup_AWS1/
ubuntu@ip-172-31-27-221:~/Rup_AWS1$ ls
'New Text Document.txt'  index.js  package-lock.json  package.json
ubuntu@ip-172-31-27-221:~/Rup_AWS1$ npm install

Run `npm audit` for details.
npm notice
npm notice New major version of npm available! 8.19.4 -> 10.5.2
npm notice Changelog: https://github.com/npm/cli/releases/tag/v10.5.2
npm notice Run npm install -g npm@10.5.2 to update!
npm notice
ubuntu@ip-172-31-27-221:~/Rup_AWS1$ npm -v
8.19.4
ubuntu@ip-172-31-27-221:~/Rup_AWS1$ node index.js
Started server

```

15. Now copy the “Public IPv4 address” & paste it on a new tab. Add “:4000” at the end and then press enter.



Therefore, the project has been successfully deployed from GitHub to EC2.